

Bachelor in Computer Engineering – Management and Information Systems

Introduction to Cybersecurity

Lab 1

Reconnaissance & Enumeration

Student Assignment

Course	Introduction to Cybersecurity
Lab	1 of 4
Topic	Reconnaissance & Enumeration
Duration	90–120 minutes
Difficulty	Beginner
Flags	3

Learning Objectives

After completing this lab you will be able to:

- Perform basic active reconnaissance on a target host using standard tools
- Identify exposed services and gather information from HTTP responses
- Use discovered credentials to obtain shell access via SSH
- Navigate a multi-hop network environment (firewall, router, server, workstation)
- Document findings in a structured penetration testing report

Scenario

You are a junior penetration tester hired to assess the perimeter of **PEBCAK Corp** – a fictional company whose name stands for *Problem Exists Between Chair And Keyboard*.

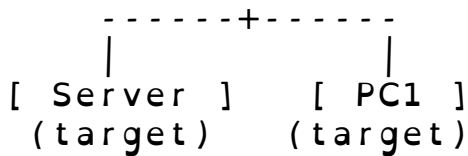
Your point of entry is the **Attacker** node in the lab environment. The target company exposes a public hostname: **Lab1**. Your goal is to enumerate the target, find exposed information, and pivot as deep as possible into the internal network.

Rules of engagement

- Only attack nodes within the lab environment
- Do not modify pfSense firewall configuration
- Document every step – commands run, output received, decisions made
- If the lab breaks, notify your instructor (do not attempt to fix infrastructure)

Network Topology

```
[ Attacker - Parrot OS ]
      |
[ Firewall - pfSense ]   <-- entry point: Lab1
      |
[ Router - VyOS ]
      |
```



Node	Role	Access
Attacker	Your machine (Parrot OS)	VNC console
Firewall	Perimeter firewall (pf-Sense)	Via exploitation
Router	Internal router (VyOS)	<i>infrastructure – do not modify</i>
Server	Target web + SSH server	Via exploitation
PC1	Internal workstation	Via exploitation

Table 1: Lab nodes overview

Entry point: The hostname `lab1` resolves to the firewall's public IP. All inbound traffic goes through the firewall before reaching internal nodes.

Tasks

Complete the following tasks in order. Document each step.

Task 1 – Web Reconnaissance

1. Open a browser on the Attacker node and navigate to `http://lab1`
2. Identify the company name displayed on the page
3. View the page source (`Ctrl+U` or `curl -s http://lab1`)
4. Look for any hidden information or comments in the HTML

Deliverable: What hint did you find in the page source?

Task 2 – Service Discovery

1. Using the hint from Task 1, find an additional resource on the web server

2. What credentials are exposed at that resource?
3. What service do the credentials grant access to?

Deliverable: Service type, host, username.

Task 3 – Initial Access (Flag 1)

1. Use the discovered credentials to connect to the target service
2. Confirm you have a shell on the target server
3. List the contents of the home directory and read the flag file

Deliverable: Contents of the flag file. Read it carefully – it contains more than just the flag.

Task 4 – Pivot to Firewall (Flag 2)

1. Using information found in Task 3, connect to the internal firewall
2. Read the flag file in the administrator's home directory

Deliverable: Flag string. What security mistake made this possible?

Task 5 – Reach the Workstation (Flag 3)

1. The firewall flag reveals a further internal host
2. From the server, connect to the internal workstation
3. Read the flag in the user's home directory

Deliverable: Flag string. How many hops from the Attacker to the workstation?

Deliverables

Submit a brief report (1–2 pages) covering:

- **Reconnaissance summary:** what did you find and how
- **Access path:** step-by-step from Attacker to each target
- **All 3 flags captured:** paste the exact flag strings
- **Reflection:** one thing you would fix as the sysadmin to stop this attack

Hints

Stuck? Hints are ordered by how much they reveal. Try each before moving to the next.

1. Web servers often have more than one page. Think about what the company name suggests.
2. `curl` and your browser both let you read page source. Comments in HTML are written between `<!--` and `-->`.
3. Once you have SSH credentials, the command is: `ssh user@hostname`
4. Read the flag file *completely* – it often contains more than the flag itself.
5. From a server inside a network, you can reach other internal hosts with `ssh user@ip`.

Tools Reference

The following tools are available on the Attacker node:

Tool	Use
<code>firefox</code>	Web browser for manual reconnaissance
<code>curl</code>	Command-line HTTP client
<code>ssh</code>	Secure Shell client
<code>nmap</code>	Network scanner (for further exploration)
<code>dig</code> / <code>nslookup</code>	DNS query tools

Table 2: Available tools on Attacker node